

Name: _____

NetID: _____

1. Suppose A and B are sets such that $A \subset B$.
 - (a) Prove that that $B = A \cup (B \setminus A)$.
 - (b) Will this hold true without the assumption $A \subset B$? If yes, provide a proof. If not, provide a counterexample.
2. Let A, B and C be three sets. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be two functions.
 - (a) Prove that if the composition $g \circ f$ is one-to-one (injective), then f is also one-to-one.
 - (b) Is the converse of the previous statement true? If your answer is yes, then provide a proof. Otherwise, provide a counterexample.
 - (c) Show that if the composition $g \circ f$ is onto (surjective), then g is also onto.
 - (d) Is the converse of the previous statement true? If your answer is yes, then provide a proof. Otherwise, provide a counterexample.
3. Suppose that $x \in \mathbb{R}$ with $x > -1$. Prove that, for every $n \in \mathbb{N}$,

$$1 + nx \leq (1 + x)^n.$$

4. Let F_n denote the n -th term of the Fibonacci sequence given by $F_1 = 1$, $F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$. Prove that

$$F_1 + F_2 + F_3 + \cdots + F_n = F_{n+2} - 1.$$

5. Given two sets A and B , the set of functions $f : A \rightarrow B$ is denoted by B^A . Show that

$$\{0, 1\}^{\mathbb{N}} = \{f : \mathbb{N} \rightarrow \{0, 1\}\}$$

is uncountable by providing a bijection with another uncountable set.